Simple PoseDetect Board Guide

Easy explanation of board connection, benchmark testing, and live pose detection

What are we doing?

You have an embedded board and a host PC. The goal is to connect them, run the human PoseDetect program on the board, and check how fast it works.

Step 1 — Switch ON the board

Power on the board, connect it to the PC using a serial cable, and open PuTTY. You can now type commands into the board's Linux terminal.

Step 2 — Connect the board and PC through Ethernet

Assign IP addresses so the PC and board can communicate:

Device	IP address
Board	192.168.1.85
PC	192.168.1.52

Step 3 — Check the connection

On your PC, run:

```
ping 192.168.1.85
```

If you receive replies, the connection is working.

Step 4 — Enter the board from your PC

```
ssh root@192.168.1.10
```

SSH lets you control the board using your PC terminal. **Note:** the document configures the board as .85 but uses .10 for SSH. Confirm the board's actual IP before running this command.

Step 5 — Check whether PoseDetect is ready

```
./zcu104_posedetect_benchmark.sh check
```

This checks whether the required model and files are present. Think of it as checking everything before starting a car.

Step 6 — Test how fast PoseDetect works

```
RUNS=3 THREADS=1 DURATION=60 \
./zcu104_posedetect_benchmark.sh baseline
```

This runs the benchmark three times, using one thread, with each run lasting 60 seconds. It measures performance such as FPS (frames per second) and processing time.

Example: 25 FPS means the program processes 25 video frames in one second.

Step 7 — See live human pose detection

On your PC:

```
ssh -X root@192.168.1.10
```

On the board:

```
cd /home/root/Vitis-AI/examples/vai_library/samples/posedetect
./test_video_posedetect sp_net 0
```

The application displays a person with lines drawn on their body. It identifies body points such as the head, shoulders, elbows, knees, and ankles. This is called Human Pose Detection.

The whole process

Power ON board → Connect PC and board → Check connection → Enter board using SSH → Check PoseDetect files → Measure FPS → Run live pose detection

Final goal

Run human pose detection on the embedded board and check how efficiently it performs.